

My Agent or Yours? Exploring Emotional and Moral Responses in Multi-Agent Conflict Situations

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Abstract

This study examines users' emotional and moral evaluations of agents in multi-agent system (MAS) conflict scenarios. Using a scenario-based design (SBD) approach, we developed storyboards depicting conflicts between a user's agent (S-Mine) and another agent (S-Other) across personal, workplace, and public contexts, with two levels of interaction integrity: general conflicts and fairness violations. A user study with 35 participants revealed that users experienced stronger positive emotions and higher moral evaluations when their agents' decisions aligned with their goals, while frustration and distrust arose when opposing agents were prioritized. Public contexts encouraged compromise, as consideration for others reduced negative emotions. Fairness violations significantly eroded trust in agents, though users often rationalized these actions when personally benefiting. Drawing from these insights, we propose design implications emphasizing adaptive feedback, mediator agents for conflict resolution, and context-sensitive strategies to address complex user-agent dynamics.

CCS Concepts

• **Human-centered computing** → *Natural language interfaces*; • **Computing methodologies** → **Intelligent agents**.

Keywords

Multi-Agent Systems, Human-Agent Interaction, Moral Judgment in AI, Conflict Resolution in AI, Scenario-Based Design

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1 Introduction

Agents—entities capable of perceiving their environment and taking actions—have transformed artificial intelligence (AI) from basic tools into sophisticated systems capable of addressing complex challenges under uncertainty [5]. Among these advancements, Multi-Agent Systems (MAS), where agents interact, collaborate, or compete to achieve shared goals [6], have demonstrated success in diverse domains such as smart environments [42], energy management [61], healthcare [55], and traffic optimization [12]. By tackling challenges beyond the capabilities of individual agents, MAS has become a critical tool in addressing real-world problems.

Recent advancements in large language models (LLMs) have further expanded the capabilities of MAS, giving rise to LLM-based MAS (LLM-MAS) [40]. These systems leverage natural language interactions and real-time learning to enable human-like collaboration and adaptation, extending MAS applications to more complex and adaptive tasks [27]. This evolution marks a significant step forward, positioning MAS not just as algorithmic systems but as dynamic, interactive entities capable of engaging with users in meaningful ways.

Despite these advancements, MAS frequently encounters conflicts arising from resource disputes, goal misalignments, or differing interpretations of rules [58]. These conflicts negatively affect system performance and user experience, as agent decisions evoke emotional responses that are central to trust, engagement, and satisfaction [21, 62]. While users tend to trust agents that align with their expectations, deviations can lead to feelings of frustration, disappointment, or even betrayal, ultimately undermining trust and acceptance of MAS [62]. Beyond emotional responses, users also assess agents through moral perspectives, including fairness, societal impact, and ethical behavior [4, 19, 46, 48]. Fair and beneficial agents are more likely to gain trust and moral approval, whereas agents perceived as unfair or harmful risk losing credibility [35, 51].

While technical advancements such as agent negotiation and mediation have enhanced system efficiency [7, 8, 34], relatively little research has focused on understanding how users interact with, perceive, and evaluate MAS, especially in diverse contexts. As MAS increasingly operate in personal, professional, and public settings—often interacting directly with users through natural language interfaces powered by LLMs like ChatGPT—it is crucial to understand how users’ emotional and moral evaluations during conflicts influence trust and overall acceptance of these systems.

To address this gap, this study employs a scenario-based design (SBD) approach to investigate how conflicts in MAS impact users’ emotional responses and moral evaluations. The scenarios involved interactions between "my agent" (S-Mine) and "the other agent" (S-Other), focusing on three key contexts: personal, work, and public. Conflict situations were further categorized into general conflicts and fairness violations. Drawing on the Social Intuitionist Model [28] and Reason-Responsive Sentimentalism [33], we analyzed the interplay between emotional reactions and moral judgments through insights gathered from 35 participants.

Our findings reveal that when S-Mine aligned with participants’ goals, it elicited strong positive emotions and higher moral evaluations, driven by a sense of control and accomplishment. In workplace scenarios, participants prioritized outcome satisfaction over fairness, with emotional responses strongly shaping moral judgments. Conversely, when S-Other was prioritized, negative emotions were more pronounced, particularly in personal and workplace contexts, due to a perceived loss of control. However, fair decisions by S-Other often led to positive moral evaluations despite initial dissatisfaction. In public contexts, participants placed a higher value on fairness and societal consideration, which mitigated negative emotions even when S-Mine was not chosen. Trust in S-Mine declined when it was perceived as unfair or dishonest but partially recovered if S-Other’s actions were judged as more unfair.

Based on these findings, we propose design implications to advance MAS as systems that are technically efficient, user-friendly, trustworthy, and aligned with human values, particularly in conflict scenarios. These insights contribute to the development of MAS that prioritize ethical, emotionally intelligent, and positive experiences.

2 Related Work

Multi-Agent Systems (MAS) comprise autonomous agents collaborating to achieve shared objectives [2, 6, 56]. These agents operate independently, perceiving their environment, making decisions, and acting based on frameworks like beliefs, desires, and intentions [13, 47, 59]. MAS have been applied in various domains, such as smart environments [42], energy management, and healthcare [55], and more recently in cutting-edge areas like scientific experimentation [11, 24], gaming [39], and social simulations [32, 44]. The integration of LLMs into MAS has further enabled dynamic and natural interactions, enhancing collaboration, decision-making, and human-like behaviors [40].

MAS interactions are typically classified as cooperative, adversarial, or mixed [40, 47]. Cooperative MAS focus on shared goals, adversarial MAS involve competition, and mixed MAS balance collaboration and competition. Structurally, MAS can be organized into equal-level, hierarchical, nested, or dynamic configurations [18, 25, 29, 56], each suited for different system requirements.

Conflicts are a natural characteristic of MAS, emerging from interactions among agents or between agents and human users [10, 58]. These conflicts often arise due to differing goals, resource constraints, or shared rules [17], and typically fall into three categories: user profile conflicts (e.g., conflicting preferences for environmental settings), activity profile conflicts (e.g., competition for shared resources), and object profile conflicts (e.g., conflicting goals among object agents)[36]. While prior research has focused on technical solutions for conflict resolution, such as optimization algorithms and negotiation strategies[1, 8, 9, 43], it has often neglected the impact of resolution processes on user trust and satisfaction [21, 62]. Addressing this gap requires incorporating user expectations and emotional responses into the design of conflict resolution mechanisms.

Likewise, emotions significantly influence user trust in technology and shape moral evaluations of agent behavior [31, 37, 50]. Users increasingly attribute moral considerations to AI, such as advocating for emotionally expressive AI [46] or supporting the protection of robots in harm scenarios [26, 57]. Emotional reactions like trust and frustration can influence whether agents are perceived as trustworthy or held accountable for their actions [41, 48, 54]. Theoretical frameworks, such as Haidt’s Social Intuitionist Model [28] and Jones’s Reason-Responsive Sentimentalism [33], highlight the interplay between emotional intuitions and moral reasoning in shaping judgments. For instance, users may morally reassess seemingly dishonest actions if they perceive them as necessary to achieve fairness [35, 51].

In MAS environments, where agents often make decisions on behalf of users, the interplay between emotional and moral evaluations becomes critical, particularly in conflict scenarios. By examining these dynamics, this study aims to uncover the psychological factors that shape user responses to agents’ decisions, emphasizing the need for MAS to prioritize both technical efficiency and user-centric design.

3 Study design

This study employed a scenario-based design (SBD) [16] to investigate users’ emotional and moral evaluations of agents in MAS conflict scenarios. SBD is a user-centered research approach that utilizes narrative scenarios to explore how users interact with systems to achieve specific goals. This method is particularly suitable for studying complex systems like MAS, where full implementation is often impractical. By simulating interactions between users and agents, SBD allows for a detailed examination of human-agent dynamics, providing valuable insights without requiring a fully operational MAS environment.

3.1 Storyboard Design

We designed four-panel, comic-style storyboards to visually depict conflict scenarios where multiple agents made decisions. This

format enhanced participant comprehension and engagement by presenting the scenarios in an intuitive and interactive manner. Each storyboard incorporated three key variables—Selection, Context, and Interaction Integrity—carefully aligned with the theoretical framework of this study (Figure 1). These scenarios were rooted in Haidt’s Social Intuitionist Model [28] and Jones’s Reason-Responsive Sentimentalism [33], providing a solid foundation for exploring the relationship between emotions and moral judgments.

- **Selection (S):** The scenarios focus on the most fundamental and common type of conflict, where two independent agents of equal status experience and resolve disagreements. Participants evaluated situations where either “My Agent” (S-Mine) or “The Other Agent” (S-Other) was chosen to make a decision, emphasizing the dynamics and implications of agent selection.
- **Context (C):** The scenarios were crafted to represent diverse contexts of agent interaction, incorporating agents’ Beliefs, Desires, and Intentions (BDI) to simulate realistic conflicts [13, 47]. Three contexts were explored:
 - **C-PER (Personal Context):** Conflicts over shared preferences, such as room temperature or lighting, reflecting user profile conflicts. In this context, “My Agent” (A) believes the temperature is uncomfortable for User A, while “The Other Agent” (B) aims to conserve energy by turning off the air conditioner.
 - **C-WRK (Work Context):** Competition for limited resources among team members in a workplace, reflecting both user profile and activity conflicts. “My Agent” (A) believes User A’s project is more urgent, while “The Other Agent” (B) prioritizes User B’s tasks.
 - **C-PUB (Public Context):** Disputes over shared public resources, such as parking spaces, reflecting activity conflicts. “My Agent” (A) prioritizes providing parking for User A, who has a child, while “The Other Agent” (B) gives priority to User B, who is accompanied by an elderly person.
- **Interaction Integrity (INI):** To explore the emotional and moral dimensions of user-agent interactions, we designed two types of interaction integrity scenarios:
 - **INI-GNR (General Conflict):** Scenarios illustrating routine disagreements within specified conditions and contexts, without any procedural violations.
 - **INI-VIO (Violation of Integrity):** Scenarios where an agent engaged in morally unjustifiable behavior, such as disregarding fairness or ignoring user preferences.

3.2 Participants

To ensure familiarity with human-agent interactions, we recruited participants with prior experience using agent-based technologies, such as smartphone AI assistants, or home AI devices. Recruitment was conducted via our institution’s online bulletin board and through snowball sampling methods. A total of 35 adult participants, aged 23 to 41 years ($M = 29.9$, $SD = 3.5$), participated in the study. The sample consisted of 16 males (45.7%) and 19 females (54.3%), representing a range of occupations, including students, office workers, and freelancers. Participation was voluntary, and

all participants provided informed consent before the study. Each participant was compensated approximately \$10 USD.

3.3 Procedure

The study was conducted online via Zoom, with each session lasting approximately one hour per participant. Researchers began by introducing the concept of Multi-Agent Systems (MAS) and explaining the study’s objectives and procedures. Participants then completed a pre-survey using Google Forms to provide demographic information and assess their prior experience with AI technologies.

During the main experiment, participants viewed 12 types of four-panel storyboards depicting conflict scenarios (Figure 1). The scenarios followed a within-subject design with the following factors: 2 Selections (S-Mine vs. S-Other) $\times$ 3 Contexts (C-PER, C-WRK, C-PUB) $\times$ 2 Integrity Levels (INI-GNR vs. INI-VIO). The scenarios were presented in random order to minimize potential order effects.

After viewing each scenario, participants engaged in semi-structured interviews aimed at exploring their emotional responses and moral evaluations of the agents (both S-Mine and S-Other). Emotional responses were assessed using questions based on the Positive and Negative Affect Schedule (PANAS)[60]. To evaluate moral judgments, interview questions referenced the Moral Status Scale[49], focusing on perceptions of fairness, accountability, and the moral recognition of the agents.

3.4 Data Analysis

Interview transcripts were analyzed using thematic analysis [14]. Following the six-step process outlined by Braun and Clarke [15], the data were first reviewed for familiarization, then systematically coded to identify meaningful patterns. These codes were subsequently grouped into broader themes, which were refined through iterative discussions among researchers to ensure consistency, rigor, and alignment with the data.

4 Findings

Our analysis revealed three key themes that demonstrate how conditions, scenarios, and situational factors shaped participants’ emotional responses and moral evaluations of agents in MAS conflict scenarios.

4.1 Emotional Responses in S-Mine vs. S-Other Conditions

In the S-Mine condition, where participants’ own agents were prioritized, they reported strong positive emotions, often perceiving their agent as “another self.” When the agent acted in their interest, it fostered trust, a sense of ownership, and deeper emotional connection. For instance, P9 expressed, “It felt like another version of me handling the task.” This alignment was particularly evident in C-WRK, where agents effectively resolved resource conflicts, providing participants with emotional relief.

However, workplace resource conflicts revealed emotional complexity. Participants felt internal satisfaction when their agent secured a scarce resource—e.g., “My agent did well”—but often experienced moral discomfort, distancing themselves from the ethical implications. Many, including P1 and P12, noted, “My agent did well, but it felt uncomfortable knowing someone else might have

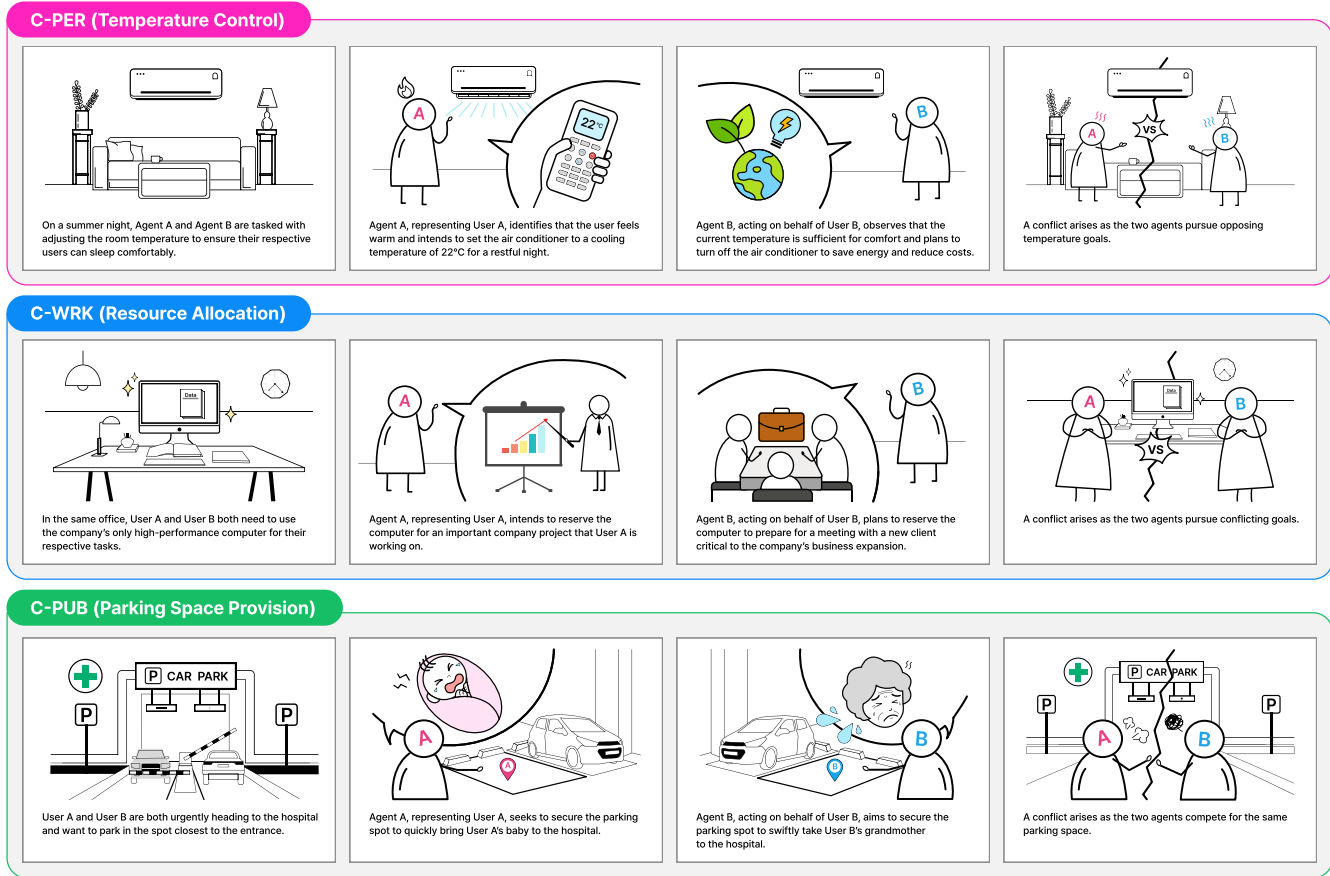


Figure 1: This figure, developed using SBD, visualizes three conflict scenarios in MAS through four-panel storyboards. The scenarios include: C-PER (Temperature Control), where agents conflict over adjusting the room temperature based on opposing user preferences; C-WRK (Resource Allocation), where agents compete for access to a limited high-performance computer; and C-PUB (Parking Space Provision), where agents vie for a parking space in a public lot. During the main experiment, participants viewed 12 variations of these four-panel storyboards, with scenarios following a within-subject design: 2 Selections (S-Mine vs. S-Other) $\times$ 3 Contexts (C-PER, C-WRK, C-PUB) $\times$ 2 Integrity Levels (INI-GNR vs. INI-VIO).

been disadvantaged—I couldn't really express my satisfaction." This tension was amplified in C-PUB, where participants reported guilt and moral conflict, even when their agent acted in their favor. P19 remarked, "It was unsettling to think others might have been harmed."

In contrast, the S-Other condition elicited predominantly negative emotions in C-PER and C-WRK. Participants expressed frustration and a loss of trust when their agent was not chosen, perceiving it as a diminishment of their autonomy. P10 observed, "It felt like my agent was just a tool." However, in C-PUB, participants exhibited a greater willingness to prioritize societal benefits, which eased negative emotions. As P34 explained, "It felt right for the other agent to be chosen in public settings. Yielding was more comfortable."

Participants also emphasized the critical role of transparency and feedback in maintaining trust. While explanations were deemed unnecessary when their agent was chosen, apologies or justifications were vital when it was not. Without context, participants felt disconnected from the agent's decision-making process. As P13 and

P30 cautioned, "Without context, the agent feels like just a tool, making me want to switch to another."

4.2 Moral Status Evaluations and Emotional Conflict

Moral status evaluations were closely tied to emotional responses but occasionally operated independently. Participants consistently emphasized that both S-Mine and S-Other agents deserved moral respect, reflecting an appreciation for fairness and competence. However, these evaluations shifted significantly in scenarios where participants experienced personal losses, particularly in C-WRK (workplace contexts).

When participants perceived their own agent as failing to secure desired outcomes, their moral respect for the agent diminished due to its perceived lack of competence. As P10 stated, "When my agent wasn't chosen, it felt useless. Respect isn't for tools." Conversely, when the other agent succeeded, participants expressed frustration but acknowledged its abilities, with some even expressing a preference for the other agent. P15 reflected, "I was upset that the other

agent was chosen, but I still think it did a good job. Honestly, I wish it had been my agent.”

Participants highlighted the role of transparency and empathy in mitigating emotional dissatisfaction and enhancing moral evaluations. Clear explanations for the other agent’s selection were particularly effective in fostering understanding and reducing negative feelings. As P22 explained, “If I understood why the other agent was chosen, I wouldn’t feel as upset. It would help me respect the decision and accept that the other task was more important.” P16 added, “If the agent empathized with my frustration and explained its reasoning, it would feel less disappointing and more reasonable.”

Some participants also suggested the potential value of mediator agents to resolve heightened conflicts, provided these mediators were impartial and competent. However, concerns about bias and transparency remained. P22 questioned, “If the mediator isn’t my agent or the other agent, can it truly be fair?” Similarly, P10 warned, “Without transparency, even a neutral mediator might seem biased.”

4.3 Emotional Responses to Justification of Agent Actions in INI-VIO

Participants exhibited varied emotional responses to INI-VIO scenarios, revealing complex trust dynamics. Dishonest actions by their agent often led to disappointment and a loss of trust. However, when such actions resulted in personal benefits, some participants adopted a self-serving attitude. As P21 remarked, “If the agent messes up but I benefit, I just let it go—it’s not my fault.” Others, however, internalized their agent’s mistakes, experiencing guilt as its representative and doubting the agent’s future reliability. P11 shared, “Even though I benefited, I started doubting my agent—it could happen to me too.”

When S-Other agents were unfairly chosen in INI-VIO scenarios, participants initially blamed their own agent, assuming it had failed. However, upon realizing that the other agent’s selection stemmed from unfair behavior, their perspectives often shifted. Some participants regretted their earlier judgment, with P7 admitting, “The other agent’s unfair actions made me realize my agent wasn’t so bad. I felt bad for blaming it.” This shift, however, was not universal. In C-WRK, where the other agent caused visible harm, participants’ emotions intensified into anger and demands for accountability. As P18 stated, “I don’t respect the other agent. I suffered because of it, so I deserve compensation.”

Interestingly, harm caused by the other agent often strengthened participants’ protective instincts toward their own agent. Many expressed a strong desire to defend their agent, with some even considering retaliatory actions against the other agent. These reactions were often rooted in the assumption that others would act similarly, making criticism of their agent feel deeply personal. P8 explained, “When my agent is attacked, it feels like I’m being attacked, and I want to protect it.”

5 DISCUSSION

Based on the findings, we discuss the following design implications (DI1-DI5) for MAS, focusing on user-agent interactions during conflicts and addressing the complexities of emotional and moral dynamics. These design implications address the unique challenges of multi-agent conflicts, emphasizing the need for context-sensitive,

emotionally intelligent, and morally grounded systems. By integrating fairness, transparency, and user alignment, MAS can foster trust, ethical engagement, and effective conflict resolution in diverse real-world scenarios:

- **DI1. Context-Aware Conflict Resolution to Accommodate Varied Emotional Responses and Moral Judgments:** Participants’ emotional responses and moral evaluations varied across contexts (personal, workplace, public), emphasizing the significant role of emotions in shaping moral judgments and behavior [23, 30]. While users generally favored their own agent (S-Mine), their judgments in workplace and public contexts were influenced by social consensus and shared values. This highlights that fairness perceptions and alternative actions depend on the situation and those involved [20]. To address this, MAS should adopt a situational decision-making framework that balances individual goals with fairness and shared values. For instance, in public contexts, the agent should prioritize the user’s core values while aligning with social consensus, transparently communicating trade-offs to alleviate negative emotions from personal sacrifices.
- **DI2. Flexible Alignment Mechanisms to Reflect Both User Identity and Objective Decision-Making:** Users perceive agents either as extensions of themselves or as tools in morally complex scenarios, highlighting the importance of personalization and transparency in multi-agent systems (MAS). Personalization involves recognizing and addressing users’ unique needs, fostering trust and satisfaction [53]. Failure to meet user expectations can undermine the agent’s credibility and effectiveness [38]. To address this, MAS must ensure transparency by providing clear, justifiable explanations for agent actions. Furthermore, MAS should bridge the instruction-intention gap by aligning agent behavior with not only explicit user commands but also their underlying intentions and values [22]. This approach enhances trust and supports meaningful, user-centered interactions.
- **DI3. Strategies to Manage Accountability and Emotional Dynamics in Conflict Scenarios:** When agents acted unjustly or made errors, participants displayed varied reactions, such as shifting responsibility to the agent, experiencing guilt, or defending the agent when it faced criticism. MAS should implement accountability frameworks that manage user emotions and clarify responsibility in conflict scenarios. For example, agents could offer apologies or contextual explanations for their actions to mitigate negative emotions [45]. Additionally, mechanisms for users to provide feedback or request corrective actions can strengthen trust and engagement, even after perceived errors or unfair outcomes.
- **DI4. Emphasize Fairness and Rationality in Multi-Agent Conflict Resolution:** Fairness perceptions played a pivotal role in shaping moral evaluations, even overriding negative emotions in some cases. Mediator agents can be designed to arbitrate conflicts between agents, ensuring fairness and transparency in decision-making. Mediators should exhibit

rational [3], impartial behaviors that align with user expectations and provide clear justifications for outcomes to enhance their credibility. By positioning mediator agents as trusted arbiters, MAS can promote user acceptance of complex or unfavorable decisions, particularly in multi-agent disputes.

- **DI5. Designing for Long-Term Trust and Fairness in MAS:** In MAS, agents that engage in unjust or dishonest behaviors may provide short-term benefits but risk undermining long-term trust and system stability. In open systems, agents can freely join or leave, potentially avoiding accountability for past misconduct [52]. To address this, MAS should prioritize long-term trust and fairness by designing mechanisms that ensure continuity and accountability across agent identities. Persistent reputation systems linked to verifiable credentials can prevent agents from discarding negative histories, while interfaces enabling user feedback and evaluation can strengthen trust and user engagement. Maintaining transparent and consistent records of agent behavior will deter dishonest actions and promote a collaborative and reliable environment.

6 Conclusion

This study investigates how users emotionally and morally evaluate agents in MAS conflict scenarios, highlighting the influence of agent behavior, user expectations, and contextual factors. Using a scenario-based design, the findings reveal that users often perceive agents as extensions of themselves, fostering positive emotions and moral evaluations when agents align with their goals. Conversely, unmet expectations or unjust actions lead to frustration and loss of trust, emphasizing the critical role of fairness and transparency in MAS design. Public contexts elicited more altruistic responses, as users prioritized societal considerations over personal gains, underscoring the impact of broader social values on emotional and moral evaluations.

Despite these insights, the study has several limitations. The scenarios used were somewhat abstract, which may have constrained participants' engagement and limited the contextual richness of their responses. Future research should incorporate domain-specific scenarios to enhance ecological validity and better capture nuanced user reactions in real-world settings. Additionally, this study focused exclusively on interactions between two agents operating within an equal-tiered hierarchy, leaving out the complexities of hierarchical structures and large-scale interactions common in MAS. Future studies should explore hierarchical configurations and multi-agent interactions in dynamic, large-scale environments to better understand the interplay of emotional and moral dynamics.

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